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ELECTRONIC DEVICE, OPERATING METHOD THEREOF, AND ELECTRONIC SYSTEM

Abstract

An electronic device includes: a memory configured to store a first training dataset labeled with a defect shape and a mask pattern of a mask and at least one artificial intelligence model; and at least one processor configured to receive defect data including first defect data, second defect data, and third defect data; by using the second defect data, obtain a second training dataset by performing data augmentation on a defect shape for which an amount of labeled data in the first training dataset is determined to be less than a threshold value; train a first artificial intelligence model generated to classify a defect type of the mask based on the first training dataset and the second training dataset; and classify a defect type of the mask based on at least one of the first defect data, the third defect data, or the first artificial intelligence model.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0030840 filed in the Korean Intellectual Property Office on Mar. 4, 2024, the entire contents of which are herein incorporated by reference.

BACKGROUND

1. Field

[0002] One or more example embodiments of the disclosure relate to an electronic device, an operating method of the electronic device, and an electronic system.

2. Description of the Related Art

[0003] In recent years, as the patterns being implemented on wafers have become finer, controlling defects that may occur on a mask has become important. In the related art, operators perform visual analysis of all defect images to detect mask defects at an early stage, but this method is time-consuming and there may be inconsistent analysis results between operators.

[0004] Therefore, an automatic defect classification process may be performed to control defects that may occur on the mask. In the related art, defect classification has been performed using methods, such as Convolutional Neural Network (CNN) or K-Nearest Neighbor (K-NN) algorithms. However, with the introduction of Extreme Ultraviolet Lithography (EUV) process, the process has become more sophisticated, requiring a more reliable automatic defect classification process.

SUMMARY

[0005] The disclosure attempts to provide an electronic device, an operating method of the electronic device, and an electronic system that automatically classify types of defects that may occur on a mask.

[0006] An embodiment of the disclosure provides an electronic device including: a communication circuit; a memory configured to store a first training dataset and at least one artificial intelligence model, the first training dataset being labeled with a defect shape and a mask pattern of a mask; and at least one processor operatively connected to the communication circuit and the memory, wherein the at least one processor is configured to receive, from an external database via the communication circuit, defect data obtained from at least one facility, the defect data including first defect data, second defect data, and third defect data; by using the second defect data, obtain a second training dataset by performing data augmentation on a defect shape for which an amount of labeled data in the first training dataset is determined to be less than a threshold value; train a first artificial intelligence model generated to classify a defect type of a mask based on the first training dataset and the second training dataset; and classify a defect type of the mask based on at least one of the first defect data, the third defect data, or the first artificial intelligence model.

[0007] Another embodiment of the disclosure provides an operating method of an electronic device, the operating method including: receiving, via a communication circuit, defect data obtained from at least one facility from an external database, the defect data including first defect

data, second defect data, and third defect data; by using the second defect data, obtaining a second training dataset by performing data augmentation on a defect shape for which an amount of labeled data in the first training dataset is determined to be less than a threshold value; training a first artificial intelligence model generated to classify a defect type of a mask based on the first training dataset and the second training dataset; and classifying a defect type of a mask based on at least one of the first defect data, the third defect data, or the first artificial intelligence model.

[0008] Still another embodiment of the disclosure provides an electronic system including: an electronic device including a processor, a memory, and a communication circuit; at least one inspection facility configured to inspect a defect related to a mask; and a semiconductor process facility configured to perform a process based on mask defect information, in which the electronic device is configured to predict a defect type of a mask based on at least one of manufacturing process data of the mask received from the at least one inspection facility or an artificial intelligence model trained to classify a defect type of the mask, and transmit defect prediction data for the predicted defect type to the semiconductor process facility, and the semiconductor process facility is configured to perform a process according to the defect prediction data.

[0009] According to an example embodiments, automated defect classification is performed by utilizing process data other than scanning electron microscope (SEM) and/or graphic design system (GDS) images, thereby improving reliability and productivity.

[0010] According to an example embodiments, sufficient data required for defect classification is secured through data augmentation, thereby improving the accuracy of defect classification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The above and other aspects, features, and advantages in the example embodiment will be more clearly understood from the following detailed description, taken in combination with the accompanying drawing.

[0012] FIG. 1 is a block diagram of an electronic device control system according to an example embodiment.

[0013] FIG. 2 is a flowchart illustrating an operating method of an electronic device according to an example embodiment.

[0014] FIG. 3 is a block diagram of a program module executed by an electronic device according to an example embodiment.

[0015] FIGS. 4 to 7 are diagrams for illustrating operations of an electronic device according to an example embodiment.

[0016] FIG. 8 is a diagram illustrating an electronic system according to an example embodiment.

[0017] FIG. 9 is a drawing illustrating an example of a computer device implementing the electronic device according to an example embodiment.

DETAILED DESCRIPTION

[0018] In the following detailed description, only certain example embodiments have been illustrated and described, simply by way of illustration. The disclosure may be variously implemented and is not limited to the following embodiments.

[0019] The drawings and description are to be regarded as illustrative in nature and not restrictive. Like reference numerals designate like elements throughout the specification.

[0020] In addition, a size and a thickness of each configuration illustrated in the drawings may be arbitrarily illustrated for understanding and ease of description, but the disclosure is not limited thereto. In the drawings, thicknesses of layers, films, panels, regions, etc., may be exaggerated for clarity. In the drawings, for understanding and ease of description, thicknesses of some layers and areas may be exaggerated.

[0021] Further, it will be understood that when an element such as a layer, film, region, or substrate is referred to as being “on” another element, it may be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present. Further, it will be understood that when an element such as a layer, film, region, or substrate is referred to as being “on” another element, it may be directly on the other element or intervening elements may also be present.

[0022] In addition, unless explicitly described to the contrary, the word “comprise”, and variations such as “comprises” or “comprising”, will be understood to imply the inclusion of stated elements but not the exclusion of any other elements.

[0023] Further, in the entire specification, when it is referred to as “in a plan view”, it means when a target part is viewed from above, and when it is referred to as “in a cross-sectional view”, it means when the cross-section obtained by cutting a target part vertically is viewed from the side.

[0024] In addition, the terms “. . . er”, “. . . or”, and “. . . module” described in the specification mean units for processing at least one function and operation and may be implemented by hardware components or software components and combinations thereof. Further, a plurality of “. . . modules,” a plurality of “. . . units,” or a plurality of “. . . modules” may be integrated into at least one module and implemented on at least one processor, except for “. . . units,” “. . . units,” and “. . . modules,” which need to be implemented on specific hardware.

[0025] In the present specification, “transmit or provide” may include not only direct transmission or provision, but also indirect transmission or provision through another device or by utilizing a bypass path.

[0026] In the present specification, expressions in the singular form may be construed as either singular or plural, unless the explicit words “one” or “single” or the like are used.

[0027] Hereinafter, an electronic device according to an example embodiment will be described with reference to FIG. 1.

[0028] FIG. 1 is a block diagram of an electronic device according to an example embodiment.

[0029] Referring to FIG. 1, an electronic device **100** according to an example embodiment may include at least one processor **110**, a memory **120**, a communication circuit **130**, and a display **140**. In some embodiments, the electronic device **100** may omit at least one of components described above (for example, the display **140**) or may additionally include other components (for example, an input device).

[0030] The at least one processor **110** may be operatively connected to the memory **120**, the communication circuit **130**, and the display **140**. The processor **110** may control the operation of the electronic device **100** by controlling at least one component of the electronic device **100** connected to the processor **110**.

[0031] The processor **110** may execute instructions stored in the memory **120**. The processor **110** may execute one or more applications stored in the memory **120**.

[0032] Each application may include a set of instructions. By executing the instructions stored in the memory **120**, the processor **110** may cause the electronic device **100** to perform the operations described herein. The operations described herein as being performed by the processor **110** may be understood to be performed by the electronic device **100**, as the operations may be performed by the processor **110** and/or by at least one component of the electronic device **100** connected to the processor **110**.

[0033] According to an example embodiment, the memory **120** may store data used or received by at least one component of the electronic device **100** (for example, the processor **110** and/or the communication circuit **130**). The memory **120** may store instructions that are executed by the at least one processor **110**. For example, the memory **120** may store information about a defect type of a mask that has been predicted (or classified) by the processor **110**. According to an example embodiment, the defect type of the mask stored in the memory **120** may serve as reference data when a labeling operation is performed by a user (for example, an operator) after the defect type is

stored.

[0034] According to an example embodiment, the memory **120** may store data that is transceived via the communication circuit **130**. For example, the memory **120** may store defect data received via the communication circuit **130**. According to an example embodiment, the defect data may be obtained from a server, a web storage, or an external storage device (for example, an external database **101**, or an external memory card) and stored in the memory **120**. Specifically, the memory **120** may store defect data received from the external database **101** (for example, facility database (facility DB)) via the communication circuit **130**.

[0035] According to an example embodiment, the memory **120** may store defect data obtained from at least one facility. According to an example embodiment, the defect data may include first defect data, second defect data, and/or third defect data obtained from a separate facility. The second defect data may be obtained from a mask from which the first defect data was obtained and at least one unit process was performed thereon. Further, the third defect data may be obtained from a mask from which the second defect data was obtained and at least one unit process was performed thereon.

[0036] For example, the memory **120** may store at least one of the first defect data obtained from a raw material (for example, quartz), the second defect data obtained from the mask on which a patterning process was performed, or the third defect data obtained from the mask on which a repair process was performed.

[0037] According to an example embodiment, the memory **120** may store one or more artificial intelligence models (or neural network models) and a set of training data. The one or more artificial intelligence models may include artificial intelligence models that perform learning by classification among various learning methods of deep learning or machine learning. For example, the one or more artificial intelligence models may include a first artificial intelligence model trained to classify the defect type of the mask using training data.

[0038] The training dataset may be a training dataset utilized to train the artificial intelligence model. For example, the training dataset may include image data labeled with labels indicative of defect shapes and mask patterns.

[0039] The communication circuit **130** may support establishing a wired or wireless communication channel between the electronic device **100** and an external electronic device, and performing communication over the established communication channel. The processor **110** may obtain defect data from a cloud server, a web storage, or an external storage device (for example, the external database **101** and an external memory card) via the communication circuit **130**.

[0040] The processor **110** may receive, via the communication circuit **130**, defect data obtained during a mask manufacturing process from the external database **101** (for example, a facility database (facility DB)). The external database **101** may store data obtained from at least one defect inspection facility. For example, the external database **101** may store at least one of the first defect data obtained from the raw material (for example, quartz), the second defect data obtained from the mask on which the patterning process has been performed, or the third defect data obtained from the mask on which the repair process has been performed.

[0041] According to an example embodiment, the external database **101** may receive the first defect data from a first inspection facility corresponding to a raw material inspection facility, the second defect data from a second inspection facility inspecting masks on which a patterning process has been performed, and the third defect data from a third inspection facility inspecting masks on which a repair process has been performed.

[0042] According to an example embodiment, the communication circuit **130** may include a wireless communication module (for example, a cellular communication module, a near field communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module (for example, a local area network (LAN) communication module, or a power line communication module).

[0043] According to an example embodiment, the processor **110** may communicate with external electronic devices via a first network (for example, a short-range communication network such as Bluetooth, WiFi direct, or infrared data association (IrDA)) and/or a second network (for example, a long-range communication network such as a cellular network, the Internet, or a computer network (for example, LAN or WAN)) by using the communication circuit **130**. Various types of communication circuit **130**, including those described above, may be implemented on a single chip, or may be implemented as separate chips, respectively.

[0044] According to an example embodiment, the display **140** may display information that is processed on the display **140** under the control of the processor **110**. For example, the display **140** may display various content (for example, text, images, video, icons, and/or symbols). According to an example embodiment, the display **140** may include a liquid crystal display (LCD), a light-emitting diode (LED) display, or an organic light-emitting diode (OLED) display.

[0045] According to an example embodiment, the display **140** may include a touch screen, for example, and may receive touch, gesture, proximity, or hovering input by using an electronic pen or a part of the user's body. In this case, the display **140** may also be used as an input device, but is not limited thereto. In some embodiments, the electronic device **100** may include a separate input device.

[0046] According to an example embodiment, the display **140** may visually present various information to a user (for example, an operator) of the electronic device **100**. According to an example embodiment, the display **140** may display content related to the user input in response to receiving the user input.

[0047] According to an example embodiment, the display **140** may display data processed by the processor **110**. According to an example embodiment, when the defect type of the mask is determined by the processor **110**, the display **140** may display information about the mask defect.

[0048] According to an example embodiment, the display **140** may display a graphical user interface (GUI) indicative of a result of the labeling. For example, the display **140** may display a graphical user interface (GUI) indicative of a result of the labeling of the first training data stored in the memory **120**.

[0049] According to an example embodiment, the display **140** may receive user input to re-label at least one of the defect shape and the mask pattern to corresponding data while the GUI representing the result of the labeling is displayed. According to an example embodiment, the display **140** may display a GUI indicative of a result of the re-labelling.

[0050] FIG. **2** is a flowchart illustrating an operating method of an electronic device according to an example embodiment.

[0051] Each of operations in FIG. **2** may be performed sequentially, but need not be performed sequentially. For example, the order of the operations may be reversed, and at least two operations may be performed in parallel. In some embodiments, some of the operations illustrated in FIG. **2** may be omitted, some operations may be combined, the order of some operations may be reversed, or other operations may be added.

[0052] Referring to FIG. **2**, in operation **210**, a processor (for example, the processor **110** of FIG. **1**) may receive defect data obtained from the at least one facility from an external database (for example, the external database **101** of FIG. **1**). The processor may receive the defect data obtained during the mask manufacturing process from the external database via a communication circuit (for example, the communication circuit **130** in FIG. **1**).

[0053] For example, the defect data may include first defect data, second defect data, and third defect data obtained from separate facilities. More specifically, the processor may receive at least one of the first defect data obtained from the raw material (for example, quartz), the second defect data obtained from the mask on which the patterning process has been performed, or the third defect data obtained from the mask on which the repair process has been performed.

[0054] According to an example embodiment, the first defect data may be received from a raw

material inspection facility, the second defect data may be received from an inspection facility of a mask on which a patterning process has been performed, and the third defect data may be received from an inspection facility of a mask on which a repair process has been performed.

[0055] In operation **220**, the processor may obtain, by using the second defect data, a second training dataset by data augmentation for a defect shape for which the amount of labeled data in the first training dataset is determined to be less than a threshold value. For example, the second defect data may include a scanning electron microscope (SEM) image and/or a graphic design system (GDS) image obtained from a mask. Hereinafter, for illustration, the second defect data is described as including an SEM image.

[0056] According to an example embodiment, the memory may store a first training dataset for training the first artificial intelligence model. Specifically, the first training dataset may include image data labelled with a defect shape and a mask pattern.

[0057] In an embodiment, the processor may determine a defect shape for which the amount of labeled data in the first training dataset is determined to be less than a threshold value. For example, the processor may determine that the amount of labeled data for a first defect shape among the data included in the first training dataset is less than the threshold value. Alternatively, for example, the processor may determine that the amount of data labeled for the first defect shape among the data included in the first training dataset is less than the amount of data labeled for defect shapes other than the first defect shape.

[0058] According to an example embodiment, the processor may perform data augmentation for the defect shape for which the amount of labeled data is determined to be less than the threshold value. For example, the processor may perform data augmentation for the first defect shape when the amount of labeled data for the first defect shape is determined to be less than the threshold value. Alternatively, for example, the processor may perform data augmentation for the first defect shape when the processor determines that the amount of data labeled for the first defect shape among the data included in the first training dataset is less than the amount of data labeled for defect shapes other than the first defect shape.

[0059] According to an example embodiment, the processor may perform data augmentation by synthesizing an image of the first defect shape for which the amount of labeled data is determined to be less than the threshold value and an image of at least one mask pattern.

[0060] According to an example embodiment, the processor may perform masking on an SEM image of the first defect shape for which the amount of labeled data is determined to be less than the threshold value. According to an example embodiment, the SEM image of the first defect shape may be obtained from a facility for inspecting the mask on which the patterning process has been performed.

[0061] According to an example embodiment, the processor may perform masking on the SEM image to extract (or obtain) an image of the first defect shape.

[0062] According to an example embodiment, the processor may perform data augmentation by synthesizing at least one mask pattern image onto the image of the first defect shape. For example, the processor may perform data augmentation by synthesizing various mask pattern images onto the defect image of the first defect shape.

[0063] According to an example embodiment, the processor may perform data augmentation on the defect image of the first defect shape to obtain an augmented image. According to an example embodiment, the processor may configure the augmented image as a second training dataset for use as training data and store the second training dataset in the memory.

[0064] Specific details regarding example embodiments of the method of obtaining, by the electronic device according to the disclosure, the second training dataset through data augmentation will be described below with reference to FIGS. 5 and 6.

[0065] In operation **230**, the processor may train a first artificial intelligence model generated to classify a defect type of the mask based on the first training dataset and the second training dataset.

[0066] According to an example embodiment, the processor may generate the first artificial intelligence model to classify the defect type of the mask by using an image dataset labeled with a defect shape and a mask pattern as training data.

[0067] According to an example embodiment, the processor may store the first artificial intelligence model generated to classify the defect type of the mask in the memory (for example, the memory **120** of FIG. **1**).

[0068] According to an example embodiment, the processor may train the first artificial intelligence model to classify a defect type of the mask by using the first training dataset labeled with the defect shape and the mask pattern and the second training dataset obtained through data augmentation as training data.

[0069] According to an example embodiment, the first artificial intelligence model may be implemented to include various types of deep learning networks. According to an example embodiment, the first artificial intelligence model may perform transfer learning using a second artificial intelligence model that is a pre-trained model. According to an example embodiment, the second artificial intelligence model may include, but is not limited to, ResNet 50, a type of convolutional neural network (CNN).

[0070] According to an example embodiment, the processor may train the first artificial intelligence model by fine-tuning weight parameters of the second artificial intelligence model pre-trained by using an image dataset (for example, ImageNet data) to the first artificial intelligence model. Specifically, the processor may fine-tune the weight parameters of the second artificial intelligence model pre-trained to classify the category of the image by using the image dataset to the first artificial intelligence model.

[0071] As described above, the electronic device according to the disclosure may improve data consistency by training an artificial intelligence model through transfer learning using a pre-trained model.

[0072] According to an example embodiment, the processor may train the first artificial intelligence model by applying a focal loss function. The focal loss function is used as a loss function to resolve data imbalance, which in the electronic device according to the disclosure may be used to resolve data imbalance due to a tendency for mask defects to occur. According to an example embodiment, the processor may use a method of giving a weight value to learning by relatively increasing a loss value for data that is difficult to be classified by applying the focal loss function during the training of the first artificial intelligence model.

[0073] According to an example embodiment, the processor may train the first artificial intelligence model by applying label-smoothing. Label smoothing is a regularization technique that smooths hard labels into soft labels to reduce overconfidence in deep learning predictions. According to an example embodiment, the processor may reduce an impact of label errors on model consistency by applying label smoothing when training the first artificial intelligence model.

[0074] In operation **240**, the processor may classify a defect type of the mask based on at least one of the first defect data, the third defect data, or the first artificial intelligence model.

[0075] According to an example embodiment, the processor may classify a defect type of the mask based on the defect data. According to an example embodiment, the processor may classify the defect type of the mask based on at least one of first defect data obtained from the raw material and the third defect data obtained from the mask on which a repair process has been performed.

[0076] According to an example embodiment, the processor may obtain information about at least one of a height of the defect and a location of the defect based on the first defect data. For example, the information about the height of the defect may include information about a layer in which the defect is located.

[0077] According to an example embodiment, based on the third defect data, the processor may obtain information about at least one of natural disappearance of the defect, a repair method, a repair facility, and post-repair defect disappearance.

[0078] According to an example embodiment, when the processor determines that it is unable to classify the defect type of the mask based on the first defect data, the processor may classify the defect type of the mask based on cleaning result information (e.g., information related to cleaning of the mask) included in the third defect data. For example, the processor may determine whether the defect in the mask is a stain defect based on the cleaning result information included in the third defect data.

[0079] According to an example embodiment, the processor may classify the defect type of the mask based on the first artificial intelligence model. According to an example embodiment, when the processor obtains image data (for example, second defect data) of the mask defect, the processor may classify the defect type of the mask by using the first artificial intelligence model trained to classify the defect type of the mask based on the first training dataset and the second training dataset.

[0080] According to an example embodiment, the processor may classify the defect type of the mask based on the first artificial intelligence model when it is determined that the defect type of the mask cannot be classified based on at least one of the first defect data or the third defect data.

[0081] FIG. 3 is a block diagram of a program module executed by an electronic device according to an example embodiment.

[0082] Referring to FIG. 3, the electronic device (for example, the electronic device **100** of FIG. 1) or the processor (for example, the processor **110** of FIG. 1) according to an example embodiment may include a defect classification model generation module **300** and a data organization module **310**. The defect classification model generation module **300** may include a labeling module **320**, a data augmentation module **330**, a model training module **340**, and a defect classification module **350**. In some embodiments, the processor may omit at least one of the above-described components or may additionally include other components.

[0083] According to an example embodiment, the defect classification model generation module **300**, the data organization module **310**, the labeling module **320**, the data augmentation module **330**, the model training module **340**, and the defect classification module **350** may represent a refinement of the functions executed by the processor **110**. The operations of the defect classification model generation module **300**, the data organization module **310**, the labeling module **320**, the data augmentation module **330**, the model learning module **340**, and the defect classification module **350** described below may be operations performed by the electronic device **100** as instructions stored in the memory **120** are executed by the processor **110**.

[0084] According to an example embodiment, the data organization module **310** may configure a data flow to make the defect data available to the defect classification model generation module **300**. More specifically, the data organization module **310** may configure the data flow to make the defect data available to the defect classification module **350** included in the defect classification model generation module **300**. For example, the data organization module **310** may identify, classify, generate, or transform the transceived data.

[0085] According to an example embodiment, the data organization module **310** may configure the data flow to make meta-parameters obtained during the mask manufacturing process available to the defect classification model generation module **300** for defect classification.

[0086] According to an example embodiment, the data organization module **310** may obtain first defect data from a raw material inspection facility. According to an example embodiment, the data organization module **310** may obtain the first defect data, in a form of a meta-parameter, from an inspection facility used for inspection of an incoming raw material.

[0087] For example, the first defect data may include defect information present in the raw material, for example, quartz. More specifically, for example, the first defect data may include information about at least one of a size of a defect present in the raw material, coordinates of a location of the defect, and a layer in which the defect is located.

[0088] According to an example embodiment, the data organization module **310** may obtain second

defect data from an inspection facility used for post-patterning inspection. For example, the second defect data may include information about at least one of a defect image (for example, an SEM image), a defect size, or coordinates of the location of the defect.

[0089] According to an example embodiment, the data organization module **310** may obtain third defect data from a repair process facility. For example, the third defect data may include information about at least one of the following: whether the defect naturally disappears, a repair method, a repair facility, or whether the defect disappears after repair.

[0090] According to an example embodiment, the data configuration module **310** may transmit a meta-parameter obtained during the mask manufacturing process to the defect classification model generation module **300**.

[0091] As described above, the electronic device according to the disclosure may improve the accuracy of defect classification by estimating the layer in which the defect is located through the meta-parameter.

[0092] According to an example embodiment, the defect classification model generation module **300** may include the labeling module **320**, the data augmentation module **330**, the model training module **340**, and the defect classification module **350**. However, the disclosure is not limited thereto, and one or more modules described above as being included in the defect classification model generation module **300** may be provided separately from the defect classification model generation module **300**.

[0093] According to an example embodiment, the labeling module **320** may generate a database with labels and store the labels in a database **301**. For example, the labels may include a defect shape and a mask pattern. According to an example embodiment, the labeling module **320** may store the labeled data in the database **301**.

[0094] According to an example embodiment, the labeling module **320** may be operatively coupled to a display (for example, the display **140** of FIG. 1). According to an example embodiment, the labeling module **320** may provide a labeling user interface (UI) to a user via the display. According to an example embodiment, the labeling module **320** may display a graphical user interface (GUI) representing the labeled data via the display. According to an example embodiment, the labeling module **320** may perform re-labeling based on a user input received while the GUI is displayed.

[0095] According to an example embodiment, the data augmentation module **330** may determine a defect shape in which the amount of labeled data is determined to be less than a threshold value, among the labeled data stored in the memory (e.g., the memory **120** of FIG. 1 and/or the database **301**). For example, the data augmentation module **330** may determine that the amount of labeled data for the first defect shape is less than the threshold value. Alternatively, for example, the data augmentation module **330** may determine that the amount of data labeled for the first defect shape is less than the amount of data labeled for defect shapes other than the first defect shape.

[0096] According to an example embodiment, the data augmentation module **330** may perform data augmentation for the defect shape for which the amount of labeled data is determined to be less than the threshold value. For example, the data augmentation module **330** may perform data augmentation for the first defect shape when the amount of labeled data for the first defect shape is determined to be less than the threshold value. Alternatively, for example, the data augmentation module **330** may perform data augmentation for the first defect shape when the amount of data labeled for the first defect shape is determined to be less than the amount of data labeled for defect shapes other than the first defect shape.

[0097] According to an example embodiment, the data augmentation module **330** may perform data augmentation by synthesizing an image of the first defect shape for which the amount of labeled data is determined to be less than the threshold value and an image of at least one mask pattern. For example, the data augmentation module **330** may obtain a large amount of data by synthesizing an image of the first defect shape for which the amount of labeled data is determined to be less than the threshold value and various mask pattern images.

[0098] According to an example embodiment, the data augmentation module **330** may store the data obtained through data augmentation in the database **301**. According to an example embodiment, the data augmentation module **330** may organize the data obtained through the data augmentation into datasets for use by the artificial intelligence model as training data and store the datasets in the database **301**.

[0099] As described above, the electronic device according to the disclosure may obtain a sufficient amount of labeled data through the data augmentation to maintain predictive performance in the event of a defect in a new mask pattern.

[0100] Furthermore, as discussed above, the electronic device according to the disclosure may generate clearer image data by performing data augmentation through image generation with masking.

[0101] According to an example embodiment, the model training module **340** may train an artificial intelligence model to classify defect types in the mask based on the data stored in the database **301**. For example, the data stored in the database **301** may include the data labeled by the labeling module **320** and/or the data generated by the data augmentation module **330**. More specifically, the data stored in the database **301** may include an image dataset labeled with defect shapes and mask patterns.

[0102] According to an example embodiment, the model training module **340** may train the artificial intelligence model to classify the defect type of the mask by using an image dataset labeled with a defect shape and a mask pattern as training data.

[0103] According to an example embodiment, the artificial intelligence model to classify the defect type of the mask may be implemented to include various types of deep learning networks. According to an example embodiment, the artificial intelligence model for classifying the defect type of the mask may perform transfer learning using a pre-trained model. According to an example embodiment, the pre-trained model may use, for example but is not limited to, ResNet 50, a type of convolutional neural network (CNN).

[0104] According to an example embodiment, the model training module **340** may train the artificial intelligence model by fine-tuning weight parameters of the pre-trained model pre-trained by using the image dataset (for example, ImageNet data) onto the artificial intelligence model for classifying defect types of the mask. Specifically, the model training module **340** may fine-tune the weight parameters of the pre-trained model that is pre-trained to classify categories of images by using the image dataset into the artificial intelligence model that classifies defect types of masks.

[0105] According to an example embodiment, the model training module **340** may train an artificial intelligence model to classify the defect type of the mask by applying a focal loss function. The focal loss function is used as a loss function to resolve data imbalance, which in the electronic device according to the disclosure may be used to resolve data imbalance due to a tendency for mask defects to occur. According to an example embodiment, the model training module **340** may use a method of giving a weight value to training by relatively increasing a loss value for data that is difficult to classify by applying the focal loss function during the training of the first artificial intelligence model.

[0106] According to an example embodiment, the model training module **340** may train an artificial intelligence model by applying label-smoothing. Label smoothing is a regularization technique that smooths hard labels into soft labels to reduce overconfidence in deep learning predictions.

According to an example embodiment, the model training module **340** may reduce an impact of label errors on model consistency by applying label smoothing when training the artificial intelligence model.

[0107] According to an example embodiment, the defect classification module **350** may predict defect types based on the data obtained from the data organization module **310**.

[0108] According to an example embodiment, the defect classification module **350** may classify (or determine) defect types by using meta-parameters (for example, the first defect data, the second

defect data, and/or the third defect data) obtained during the mask manufacturing process in addition to the second defect data (for example, SEM images).

[0109] According to an example embodiment, the defect classification module **350** may identify at least one of height information of a defect in the mask or location information of a defect in the mask based on the first defect data obtained from the data organization module **310**. According to an example embodiment, the defect classification module **350** may predict a defect type based on at least one of the height information of the defect in the mask or the location information of the defect in the mask.

[0110] According to an example embodiment, the defect classification module **350** may identify cleaning result information of the mask based on the third defect data obtained from the data organization module **310**. For example, the cleaning result information may include at least one of facility environment information or repair history related to the mask.

[0111] According to an example embodiment, the defect classification module **350** may predict a defect type based on the cleaning result information of the mask. For example, the defect classification module **350** may determine whether the defect on the mask is a stain based on the cleaning result information of the mask.

[0112] According to an example embodiment, the defect classification module **350** may classify (or predict) the defect type of the mask based on the artificial intelligence model trained by the model training module **340**. According to an example embodiment, the defect classification module **350** may classify the defect type of the mask based on the artificial intelligence model trained by the model learning module **340** when it is determined that the defect type of the mask cannot be classified based on at least one of the first defect data or the third defect data.

[0113] According to an example embodiment, the defect classification module **350** may store result data according to the classification of the defect type of the mask in the database **301**. According to an example embodiment, the data stored in the database **301** by the defect classification module **350** may be target data that is later labeled by a user.

[0114] As described above, the electronic device according to the disclosure may improve a speed of a defect prediction model (or defect classification model) by classifying defect types by using meta-parameters obtained during the mask manufacturing process in addition to SEM images.

[0115] As described above, the electronic device according to the disclosure may improve reliability in classifying defect types by using meta-parameters obtained during the mask manufacturing process in addition to SEM images, which reflect advanced mask manufacturing process information.

[0116] FIGS. **4** to **7** are diagrams for illustrating operations of an electronic device according to an example embodiment.

[0117] Referring to FIG. **4**, in the electronic device according to an example embodiment, the data organization module **310** may configure a data flow to make defect data available to the defect classification module **350**. In the following embodiments, duplicative descriptions of those previously described with reference to FIGS. **1** to **3** will be omitted, and only differences will be described in detail. The same reference numerals may be given to configurations identical to those of the electronic device according to an example embodiment of the disclosure described above.

[0118] According to an example embodiment, the data organization module **310** may configure data flows to make defect data available to the defect classification module **350**. For example, the data organization module **310** may identify, classify, generate, or transform the transceived data.

[0119] According to an example embodiment, the data organization module **310** may configure the data flow to make meta-parameters obtained during the mask manufacturing process available to the defect classification module **350** for defect classification.

[0120] According to an example embodiment, the data organization module **310** may obtain the first defect data in a form of a meta-parameter from a first inspection facility **410** used in incoming inspection of a raw material **401**. For example, the first inspection facility **410** may detect a defect

present in the raw material **401**.

[0121] For example, the first defect data may include defect information present in the raw material **401**, for example, quartz. More specifically, for example, the first defect data may include information about at least one of a size of a defect present in the raw material **401**, coordinates of a location of the defect, and a layer in which the defect is located.

[0122] According to an example embodiment, the data organization module **310** may obtain the second defect data in a form of a meta-parameter from a second inspection facility **420** that inspects a mask on which a first process **415** has been performed. For example, the first process **415** may correspond to a patterning process. For example, the second inspection facility **420** may detect a defect present in a mask on which the first process **415** has been performed. For example, the second defect data may include information about at least one of a defect image (for example, SEM image), a defect size, and coordinates of a location of the defect.

[0123] According to an example embodiment, the data organization module **310** may obtain the third defect data in a form of a meta-parameter from a third inspection facility **430** that inspects a mask on which a second process **425** has been performed. For example, the second process **425** may correspond to a repair process. For example, the third inspection facility **430** may detect a defect present in the mask on which the second process **425** has been performed. For example, the third defect data may include information about at least one of the following: whether the defect naturally disappears, a repair method, a repair facility, or whether the defect disappears after repair.

[0124] According to an example embodiment, the data organization module **310** may transmit meta-parameters (for example, the first defect data, the second defect data, and the third defect data) obtained during a mask manufacturing process of a mask **402** to the defect classification module **350**.

[0125] Referring to FIG. 5, in the electronic device according to an example embodiment, a data augmentation module (for example, the data augmentation module **330** of FIG. 3) may perform data augmentation for a specific defect shape. In the following embodiments, duplicative descriptions of those previously described with reference to FIGS. 1 to 3 will be omitted, and only differences will be described in detail. The same reference numerals may be given to configurations identical to those of the electronic device according to an example embodiment of the disclosure described above.

[0126] According to an example embodiment, the data augmentation module **330** may perform masking on an SEM image **510** of a first defect shape for which the amount of labeled data is determined to be less than a threshold value. According to an example embodiment, the SEM image **510** of the first defect shape may be obtained from a facility for inspecting the mask on which the patterning process has been performed.

[0127] According to an example embodiment, the data augmentation module **330** may perform masking on the SEM image **510** to extract (or obtain) a defect image **520** of the first defect shape.

[0128] According to an example embodiment, the data augmentation module **330** may perform data augmentation by synthesizing at least one mask pattern image **530** onto the defect image **520** of the first defect shape. For example, the data augmentation module **330** may perform data augmentation by synthesizing various mask pattern images **530** onto the defect image **520** of the first defect shape.

[0129] According to an example embodiment, the data augmentation module **330** may perform data augmentation on the defect image **520** of the first defect shape to obtain an augmented image **540**. According to an example embodiment, the data augmentation module **330** may store the augmented image **540** in a database (for example, the database **301** of FIG. 3) so that a model training module (for example, the model training module **340** of FIG. 3) may utilize the augmented image **540** as training data.

[0130] As described above, the electronic device according to the disclosure may improve consistency of the learning model by obtaining a sufficient amount of labeled data through data

augmentation.

[0131] Furthermore, as discussed above, the electronic device according to the disclosure may generate clearer image data by performing data augmentation through image generation with masking.

[0132] Referring to FIG. 6, in the electronic device according to an example embodiment, a data augmentation module (for example, the data augmentation module **330** of FIG. 3) may perform data augmentation by a method of synthesizing at least one mask pattern image onto a specific defect shape. In the following embodiments, duplicative descriptions of those previously described with reference to FIGS. 1 to 3 will be omitted, and only differences will be described in detail. The same reference numerals may be given to configurations identical to those of the electronic device according to an example embodiment of the disclosure described above.

[0133] According to an example embodiment, the data augmentation module may obtain a difference image **630** by subtracting an SEM image **610** of the first defect shape for which the amount of labeled data is determined to be less than a threshold value from a normal pattern image **620**. For example, the normal pattern image **620** may correspond to an image in which no defect is present.

[0134] According to an example embodiment, the data augmentation module may multiply (or synthesize) a defect mask generated based on the difference image **630** with the SEM image **610** of the first defect shape to generate a defect image **640** of the first defect shape.

[0135] According to an example embodiment, the data augmentation module may synthesize the defect image **640** of the first defect shape and at least one mask pattern image **650** to generate augmented data **660**. For example, the data augmentation module may generate the augmented data **660** by synthesizing various mask pattern images to the defect image **640** of the first defect shape.

[0136] As described above, the electronic device according to the disclosure may augment data by synthesizing various mask pattern images to the same defect image to obtain a sufficient amount of data required for defect classification.

[0137] Referring to FIG. 7, in the electronic device according to an example embodiment, a defect classification module (for example, defect classification module **350** of FIG. 3) may classify (or determine) a defect type of the mask based on at least one of manufacturing process data **701** and an SEM image **702** obtained during mask manufacturing. In the disclosure, the manufacturing process data **701** may include first defect data and third defect data, and the SEM image **702** may correspond to second defect data.

[0138] In the following embodiments, duplicative descriptions of those previously described with reference to FIGS. 1 to 3 will be omitted, and only differences will be described in detail. The same reference numerals may be given to configurations identical to those of the electronic device according to an example embodiment of the disclosure described above.

[0139] According to an example embodiment, the defect classification module may include a first classifier **710**, a second classifier **720**, an artificial intelligence model **730**, and a third classifier **740**. According to an example embodiment, the defect classification module may classify a defect type by using at least one of the first classifier **710**, the second classifier **720**, and the third classifier **740**.

[0140] According to an example embodiment, the defect classification module may obtain manufacturing process data **701** and/or an SEM image **702** that are obtained during the mask manufacturing process.

[0141] According to an example embodiment, the defect classification module may predict a defect type based on the manufacturing process data **701** (for example, the first defect data, the second defect data, and/or the third defect data).

[0142] According to an example embodiment, the first classifier **710** may identify at least one of height information of the defect in the mask or location information of the defect in the mask based on the first defect data (for example, raw material inspection results) in the manufacturing process

data **701**. According to an example embodiment, the first classifier **710** may predict the type of defect occurring in a multi-layer based on at least one of the height information of the defect in the mask or the location information of the defect in the mask. According to an example embodiment, the first classifier **710** may generate defect prediction data **750** based on the defect type predicted based on the first defect data.

[0143] According to an example embodiment, the second classifier **720** may identify cleaning result information (for example, facility environment information, and repair history) of the mask based on the third defect data in the manufacturing process data **701**. According to an example embodiment, the second classifier **720** may predict a defect type based on the cleaning result information of the mask. For example, the second classifier **720** may determine whether the defect on the mask is a stain based on the cleaning result information of the mask. According to an example embodiment, the second classifier **720** may predict the defect type of the mask using the third defect data, based on a determination that the defect type of the mask cannot be classified based on the first defect data. According to an example embodiment, the second classifier **720** may generate defect prediction data **750** based on the defect type predicted based on the third defect data.

[0144] According to an example embodiment, the third classifier **740** may classify the defect type of the mask based on the artificial intelligence model **730**. According to an example embodiment, the artificial intelligence model **730** may classify (or predict) the defect type of the mask by using the SEM image **702**. According to an example embodiment, the third classifier **740** may classify the defect type of the mask based on the artificial intelligence model **730** when it is determined that the defect type of the mask cannot be classified based on the manufacturing process data **701** (e.g., at least one of the first defect data or the third defect data). The artificial intelligence model **730** may correspond to an artificial intelligence model trained to classify defects in the mask by using a dataset labeled with defect shapes and mask patterns.

[0145] According to an example embodiment, the defect classification module may store the defect prediction data **750** predicting the defect type of the mask in a database (for example, the database **301** of FIG. 3). According to an example embodiment, the defect prediction data **750** stored in the database may be target data that is later labeled by a user.

[0146] As described above, the electronic device according to the disclosure may improve the speed of defect classification by classifying defect types by using meta-parameters obtained during the mask manufacturing process in addition to SEM images.

[0147] As described above, the electronic device according to the disclosure may improve reliability in classifying defects by reflecting advanced mask manufacturing process information (for example, facility environment information and repair history) in addition to SEM images.

[0148] FIG. 8 is a diagram illustrating an electronic system according to an example embodiment.

[0149] Referring to FIG. 8, an electronic system **800** according to an example embodiment may include an electronic device **100**, at least one inspection facility **810**, and a semiconductor process facility **820**.

[0150] According to an example embodiment, the electronic device **100** may be wirely or wirelessly connected to a semiconductor facility (for example, the inspection facility **810** and the semiconductor process facility **820**). For example, the electronic device **100** may be wirely or wirelessly connected to the semiconductor facility via a communication circuit (for example, the communication circuit **130** of FIG. 1).

[0151] According to an example embodiment, at least one inspection facility **810** may include at least one inspection facility for detecting mask defects. According to an example embodiment, at least one inspection facility **810** may include a facility for detecting defects in raw materials or in masks on which at least one process has been performed. For example, the at least one inspection facility **810** may include at least one of an inspection facility for detecting defects in raw materials, an inspection facility for detecting defects in masks on which a patterning process has been

performed, or an inspection facility for detecting defects in masks on which a repair process has been performed. According to an example embodiment, the at least one inspection facility **810** may be in wireless communication with the electronic device **100**.

[0152] According to an example embodiment, the electronic device **100** may generate defect prediction data **801** based on at least one of defect data obtained from the at least one inspection facility **810** and an artificial intelligence model trained to classify a defect type of the mask. For example, the electronic device **100** may generate the defect prediction data **801** indicating that, among A defect type, B defect type, C defect type, and D defect type, the defect type of the mask is determined to be B defect type. According to an example embodiment, the electronic device **100** may transmit the generated defect prediction data **801** to the semiconductor process facility **820**.

[0153] According to an example embodiment, the semiconductor process facility **820** may communicate wirely or wirelessly with the electronic device **100**. For example, the semiconductor process facility **820** may receive defect prediction data **801** from the electronic device **100**. According to an example embodiment, the semiconductor process facility **820** may automatically perform a repair process based on the received defect prediction data **801**. According to an example embodiment, the semiconductor process facility **820** may select a repair process based on the received defect prediction data **801**. According to another embodiment, the semiconductor process facility **820** may determine a non-defect and select releasing of a product based on the received defect prediction data **801**.

[0154] As described above, the electronic system according to the disclosure may automate the repair process based on the defect type of the mask by associating the result of the determination of the defect type of the mask with the semiconductor process facility.

[0155] FIG. **9** is a drawing illustrating an example of a computer device implementing an electronic device according to an example embodiment. The electronic device **100** of FIG. **1** may be implemented by a computer device **900** illustrated in FIG. **9**.

[0156] Referring to FIG. **9**, the computer device **900** may include a memory **910**, a processor **920**, a communication interface **930**, and an input/output interface **940**.

[0157] The memory **910** is a computer-readable recording medium, which may include a random access memory (RAM), a read only memory (ROM), and a permanent mass storage device such as a disk drive. Additionally, the memory **910** may store an operating system and at least one program code. These software components may be loaded into the memory **910** from a computer-readable recording medium that is separate from the memory **910**. The separate computer-readable recording medium may include a computer-readable recording medium, such as a hard disk, flash memory, optical disk, external hard disk, or the like. Additionally, these software components may be loaded into the memory **910** via the communication interface **930**.

[0158] The processor **920** may be configured to process instructions from a computer program by performing basic arithmetic, logic, and input/output operations. The instructions may be provided to the processor **920** by the memory **910** or by the communication interface **930**.

[0159] The communication interface **930** may provide functionality for the computer device **900** to communicate with other devices and with each other over the network **1000**. Communication methods are not limited and may include short-range wireless communication between devices as well as communication methods utilizing a communication network (for example, a cellular network, wired internet, wireless internet, or broadcast network) that the network **1000** may include. For example, the network **1000** may include any one or more of networks, such as a personal area network (PAN), a local area network (LAN), a campus area network (CAN), a metropolitan area network (MAN), a wide area network (WAN), a broadband network (BBN), and the Internet. The network **1000** may also include any one or more of network topology including, but not limited to, a bus network, a star network, a ring network, a mesh network, a star-bus network, a tree or hierarchical network, and the like.

[0160] The input/output interface **940** may serve as an interface that may transmit instructions or

data input from the user or an input/output device 950 to other component(s) of the computer device 900. In addition, the input/output interface 940 may output commands or data received from the other component(s) of the computer device 900 to the user or the input/output device 950. For example, the input/output device 950 may include an input device, such as a microphone, a keyboard, or a mouse, and an output device, such as a display or a speaker.

[0161] The example embodiments described above may be implemented in the form of computer programs that may be executable by various components on a computer, and such programs may be recorded on a computer-readable medium. The medium may include magnetic media such as hard disks, floppy disks, and magnetic tapes, optical recording media such as CD-ROMs and DVDs, magneto-optical media such as floptical disks, and hardware devices specifically configured to store and execute program instructions, such as ROM, RAM, flash memory, and the like.

[0162] The operations configuring the method according to an example embodiment may be performed in any order, provided that no order is expressly stated or implied. The order in which the operations are described does not necessarily limit the disclosure.

[0163] All examples or use of terms in this specification are intended merely to illustrate the disclosure in detail and are not intended to limit the scope of the disclosure. Further, those skilled in the art may recognize that various modifications, combinations, and changes may be made within the scope of the patent claims or equivalents thereof.

[0164] At least one of the components, elements, modules and units (collectively “components” in this paragraph) represented by a block or an equivalent indication in the drawings described above may use a direct circuit structure, such as a memory, a processor, a logic circuit, a look-up table, etc. that may execute the respective functions through controls of one or more microprocessors or other control apparatuses. Alternatively or additionally, at least one of these components may be specifically embodied by a module, a program, or a part of code, which is stored in an internal memory or an external memory of the electronic device, and contains one or more executable instructions for performing the above-described functions, and executed by one or more microprocessors or other control apparatuses. Further, at least one of these components may include or may be implemented by a processor such as a central processing unit (CPU), graphic processing unit (GPU), another type of microprocessor, or the like that performs the above-described functions. Two or more of these components may be combined into one single component which performs all operations or functions of the combined two or more components. Also, at least part of functions of at least one of these components may be performed by another of these components. Functional aspects of the above example embodiments may be implemented in algorithms that execute on one or more processors.

[0165] Although an example embodiment(s) has been described in detail, the scope of the disclosure is not limited by the example embodiment(s). Various changes and modifications using the basic concept of the disclosure defined in the accompanying claims by those skilled in the art shall be construed to belong to the scope of the disclosure.

Claims

1. An electronic device comprising: a communication circuit; a memory configured to store a first training dataset and at least one artificial intelligence model, the first training dataset being labeled with a defect shape and a mask pattern of a mask; and at least one processor operatively connected to the communication circuit and the memory, wherein the at least one processor is configured to: receive, from an external database via the communication circuit, defect data obtained from at least one facility, the defect data including first defect data, second defect data, and third defect data; by using the second defect data, obtain a second training dataset by performing data augmentation on a defect shape for which an amount of labeled data in the first training dataset is determined to be less than a threshold value; train a first artificial intelligence model generated to classify a defect

type of a mask based on the first training dataset and the second training dataset; and classify a defect type of a mask based on at least one of the first defect data, the third defect data, or the first artificial intelligence model.

2. The electronic device of claim 1, wherein the at least one processor is configured to classify the defect type of the mask based on the first artificial intelligence model, based on a determination that the defect type of the mask fails to be classified based on at least one of the first defect data or the third defect data.

3. The electronic device of claim 1, wherein the second defect data is obtained from a mask from which the first defect data was obtained and on which at least one unit process was performed, and wherein the third defect data is obtained from a mask from which the second defect data was obtained and on which at least one unit process was performed.

4. The electronic device of claim 1, wherein the at least one processor is configured to classify the defect type of the mask based on at least one of height information of a defect of the mask or location information of the defect of the mask, the at least one of the height information or the location information being included in the first defect data.

5. The electronic device of claim 3, wherein the at least one processor is configured to classify the defect type of the mask based on the third defect data, based on a determination that the defect type of the mask fails to be classified based on the first defect data.

6. The electronic device of claim 1, further comprising: a display, wherein the at least one processor is configured to: control the display to display a graphical user interface (GUI) representing labeled data through the display, and perform re-labeling based on a user input received while the GUI is displayed.

7. The electronic device of claim 6, wherein the at least one processor is configured to update the first artificial intelligence model based on a relabeled dataset based on a determination that the defect type of the mask classified by using the first artificial intelligence model is different from a defect type pre-stored for the mask.

8. The electronic device of claim 1, wherein the at least one processor is further configured to obtain the second training dataset by performing data augmentation that synthesizes an image of the defect shape, for which the amount of labeled data is determined to be less than the threshold value, and an image of at least one mask pattern.

9. The electronic device of claim 1, wherein the at least one processor is configured to train the first artificial intelligence model by fine-tuning a weight parameter of a second artificial intelligence model, which is pre-trained by using an image dataset, to the first artificial intelligence model.

10. The electronic device of claim 1, wherein the at least one processor is configured to train the first artificial intelligence model by applying a focal loss function.

11. The electronic device of claim 1, wherein the at least one processor is configured to train the first artificial intelligence model by applying label-smoothing.

12. An operating method of an electronic device including a communication circuit, a memory configured to store a first training dataset labeled with a defect shape and a mask pattern of a mask, and at least one processor operatively connected to the communication circuit and the memory, the operating method comprising: receiving, via the communication circuit, defect data obtained from at least one facility from an external database, the defect data including first defect data, second defect data, and third defect data; by using the second defect data, obtaining a second training dataset by performing data augmentation on a defect shape for which an amount of labeled data in the first training dataset is determined to be less than a threshold value; training a first artificial intelligence model generated to classify a defect type of a mask based on the first training dataset and the second training dataset; and classifying a defect type of a mask based on at least one of the first defect data, the third defect data, or the first artificial intelligence model.

13. The operating method of claim 12, wherein the classifying includes: classifying the defect type of the mask based on the first artificial intelligence model, based on a determination that the defect

type of the mask fails to be classified based on at least one of the first defect data or the third defect data.

14. The operating method of claim 12, wherein the second defect data is obtained from a mask from which the first defect data was obtained and on which at least one unit process was performed, and wherein the third defect data is obtained from a mask from which the second defect data was obtained and on which at least one unit process was performed.

15. The operating method of claim 12, wherein the classifying includes: classifying the defect type of the mask based on at least one of height information of a defect of the mask or location information of the defect of the mask, the at least one of the height information or the location information being included in the first defect data.

16. The operating method of claim 14, wherein the classifying of the defect type of the mask includes: classifying the defect type of the mask based on the third defect data, based on a determination that the defect type of the mask fails to be classified based on the first defect data.

17. The operating method of claim 12, wherein the training includes: training the first artificial intelligence model by fine-tuning a weight parameter of a second artificial intelligence model, which is pre-trained by using an image dataset, to the first artificial intelligence model.

18. The operating method of claim 12, wherein the training includes: training the first artificial intelligence model by applying a focal loss function.

19. The operating method of claim 12, wherein the training includes: training the first artificial intelligence model by applying label-smoothing.

20. An electronic system comprising: an electronic device including a processor, a memory, and a communication circuit; at least one inspection facility configured to inspect a defect related to a mask; and a semiconductor process facility configured to process a process based on mask defect information, wherein the electronic device is configured to predict a defect type of the mask based on at least one of manufacturing process data of the mask, received from the at least one inspection facility, or an artificial intelligence model trained to classify a defect type of the mask, and transmit defect prediction data for the predicted defect type to the semiconductor process facility, and wherein the semiconductor process facility is configured to perform a process according to the defect prediction data.
